

Sahil Pavaskar

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Summary

Security Engineer with 2+ years conducting security design reviews, threat modeling, and building security automation across cloud infrastructure and software systems. Skilled in risk assessments, identity management, and applied cryptography — with a track record of integrating security controls into fast-moving engineering teams and translating findings into clear security documentation for engineering and product stakeholders.

Experience

Security Software Engineer

Sep 2022 - Aug 2024

Deven Infotech Pvt Ltd

- Led security design reviews and threat modeling (STRIDE) across 8 microservices and internal automation tools, catching authentication flaws, weak cryptographic configurations, and insecure deployment practices before production.
- Built Python and Java tooling in GitHub Actions CI/CD to automate detection of exposed secrets, insecure dependencies, and identity management misconfigurations — adopted across a 12-person engineering org.
- Tuned SAST rules to cut false positives by 40%, producing secure coding guidelines that increased developer adoption of security best practices across all product teams.
- Conducted risk assessments on internal web apps, identifying auth bypass, SSRF, and injection vulnerabilities; documented business impact and tracked all fixes to closure.
- Maintained a security documentation library — threat models, risk assessment templates, and remediation playbooks — standardizing how findings were reported and resolved.

Cloud Security Engineering Intern

Feb 2021 - Aug 2021

BKC Health Center

- Implemented security controls for a hybrid-cloud healthcare environment using Terraform: IAM least-privilege policies, network security groups, TLS cipher suite enforcement, and secure system deployment configurations across AWS.
- Wrote Python scripts to continuously scan cloud assets for misconfigurations, cutting manual security assessment time by 30% and surfacing risks before production deployments.
- Investigated security incidents involving DNS misconfigurations and container network exposures; produced risk assessments with hardening steps for Docker and Kubernetes infrastructure.

Projects

AWS GuardDuty Automated Incident Response - *Python, Terraform, Lambda, EventBridge, SNS, CloudWatch*

- Built a serverless AWS pipeline that triages GuardDuty findings, isolates compromised instances via Lambda, and alerts through SNS and CloudWatch — zero manual steps in the response chain.
- Simulated credential exfiltration and lateral movement scenarios to validate detection coverage, refining alerting thresholds and producing a documented runbook for each threat category.

TLS & Cryptographic Control Audit Tool - *Python, OpenSSL, AWS, Bash*

- Built a Python tool to audit services for weak cipher suites, expired certificates, and insecure TLS versions — generating structured risk reports with remediation steps for each finding.
- Validated against a simulated microservices environment with intentionally misconfigured cryptographic controls, catching 100% of injected vulnerabilities and producing security documentation for engineering teams.

AI-Powered Code Security Reviewer - *Python, Gemini API, Bandit, GitHub Actions*

- Combined static analysis (Bandit) with LLM review (Gemini API) to automate code-level vulnerability discovery — output includes severity ratings, CWE mappings, and remediation steps consumable by non-security engineers.

Technical Skills

Languages: Python, Java, Go, C/C++, Bash

Security Domains: Threat Modeling, Risk Assessments, Security Design Reviews, Vulnerability Discovery, Secure Coding, Identity Management, Applied Cryptography

Security Tooling: SAST/DAST, Bandit, Semgrep, Burp Suite, GuardDuty, OpenSSL, custom Python automation

Protocols & Networking: TCP/IP, HTTP, DNS, TLS/SSL, OAuth 2.0, SAML, network security architecture

Infrastructure: AWS (IAM, GuardDuty, Lambda, VPC, CloudWatch, SNS), Docker, Kubernetes, Terraform, GitHub Actions

Education

California State University East Bay

Expected May 2026

Master of Science in Computer Science

Hayward, CA

Certifications

OSCP+ (Expected June 2026) | AWS Certified Solutions Architect – Associate (April 2026) | CompTIA Security+ (February 2025)